Plotting functions with GnuPlot in 2D

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1. Lab 1 - Plotting Functions

1.1 Intialisation

Windows Type `wgnuplot` in desired directory in the terminal window.

Mac OS/Linux Does not have any dedicated GUI like `wgnuplot.exe` for Windows. Type `gnuplot` in termnal to begin.

Initialising in a Windows machine:

```
G N U P L O T
       Version 6.0 patchlevel 4    last modified 2025-12-18

       Copyright (C) 1986-1993, 1998, 2004, 2007-2025
       Thomas Williams, Colin Kelley and many others

       gnuplot home:     http://www.gnuplot.info
       faq, bugs, etc:   type "help FAQ"
       immediate help:   type "help"  (plot window: hit 'h')

       Terminal type is now windows
gnuplot>
```

Note: If mathematical symbols not rendered correctly, you must run `set term win enhanced` to be sure.

1.2 Plotting basic curves

```
       gnuplot> plot sin(1/x)
```

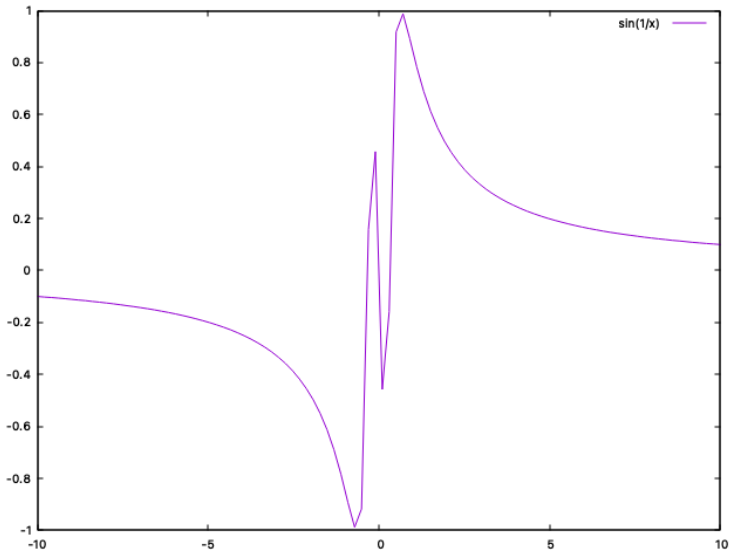


Figure 1: $\sin\left(\frac{1}{x}\right)$

1.3 Setting range and axis

```
gnuplot> set zeroaxis
gnuplot> plot [-pi:pi] sin(1/x**2)
```

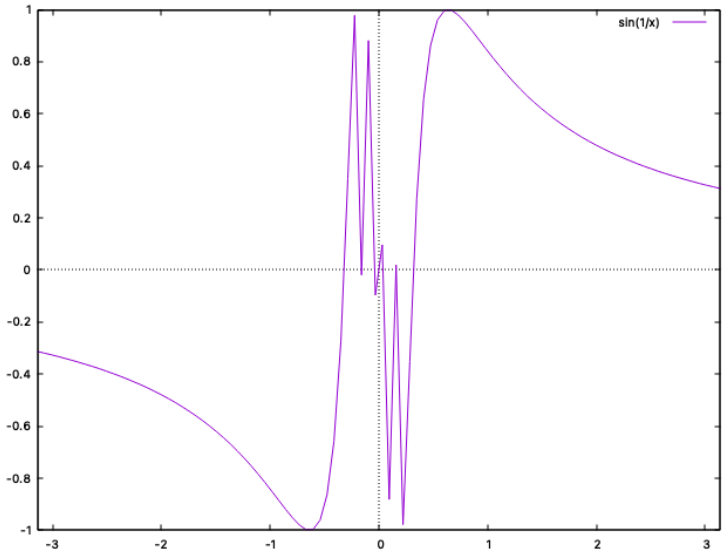


Figure 2: Range specified and zero axis made visible

1.4 Adding colours and changing line widths

```
gnuplot> plot [-pi:pi] [:] sin(x) ls 2 lw 3
gnuplot> plot [-pi:pi] [:] sin(x) ls 2 lw 3, cos(x) ls
 ↪  7 lw 3
gnuplot> plot [-pi:pi] [:] sin(x) ls 2 lw 3, cos(x) ls
 ↪  7 lw 3 lc 'blue'
```

The part `[:]` lets the y-axis autoscale. You can stack subsequent functions in one graph by separating the instructions with a `,`.

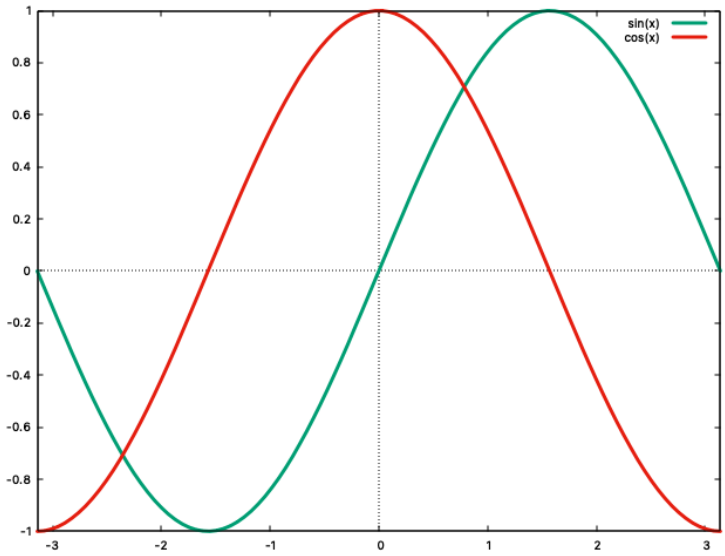


Figure 3: Default Linestyle 2 and 7 respectively for $sinx$ and $cosx$

`ls` stands for linestyle (one of the presets gnuplot comes with, different presets have different colour or dash pattern combos). However you can explicitely force features on the linestyles too for example, `lc 'blue'` forces linecolour blue on whatever colour linestyle 7 would have defaulted to.

`lw` stands for linewidth and determines how bold the lines are.

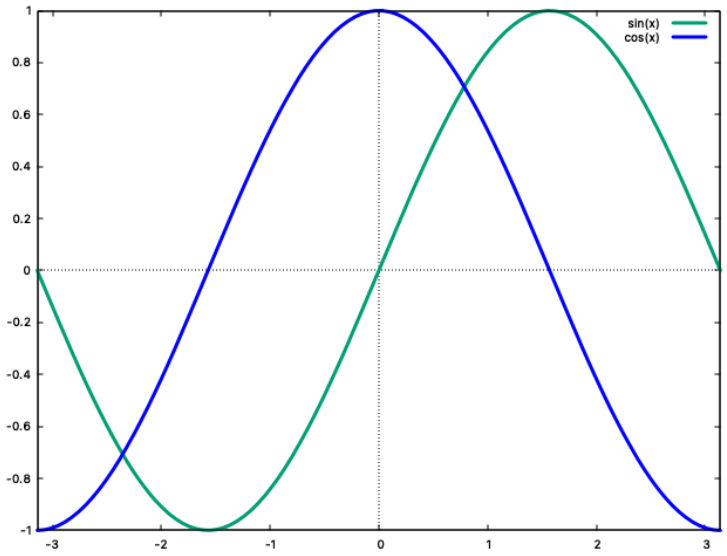


Figure 4: Forcing the colour blue on the linestyle 7 with def red

1.5 Other features

- There are different terminal types in gnuplot: some open an interactive window, some write straight to a file.
- Zoom in on functions by clicking and dragging a rectangle on the plot.
- Moving the cursor at different points on curve reveals the coordinates.
- Press ‘r’ to introduce a ruler element which is intersection of two lines and tells you the distance of different other points you hover from that given point.
- In mathematical operations use ** for exponents.

```
gnuplot> h(x) = x**2 - 2**x + 1
gnuplot> plot h(x)
```